

Network Security Report

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Introduction

This report explains the main security features I set up in the Cisco Packet Tracer network for Dublin, London and Beijing. The goal was to keep the network working normally while making it harder for the wrong users or devices to get access. I used a few different layers of security instead of relying on one feature. These included device hardening, SSH and AAA, a site-to-site VPN, OSPF authentication, ACLs, NTP and Syslog, Layer 2 switch security, DHCP snooping and PAT. Dublin contains the Sales and Purchasing networks, London contains the main servers and Admin network, and Beijing acts as the remote site. This made it easier to give each part of the network only the access it actually needed.

Network Device Hardening and Remote Access

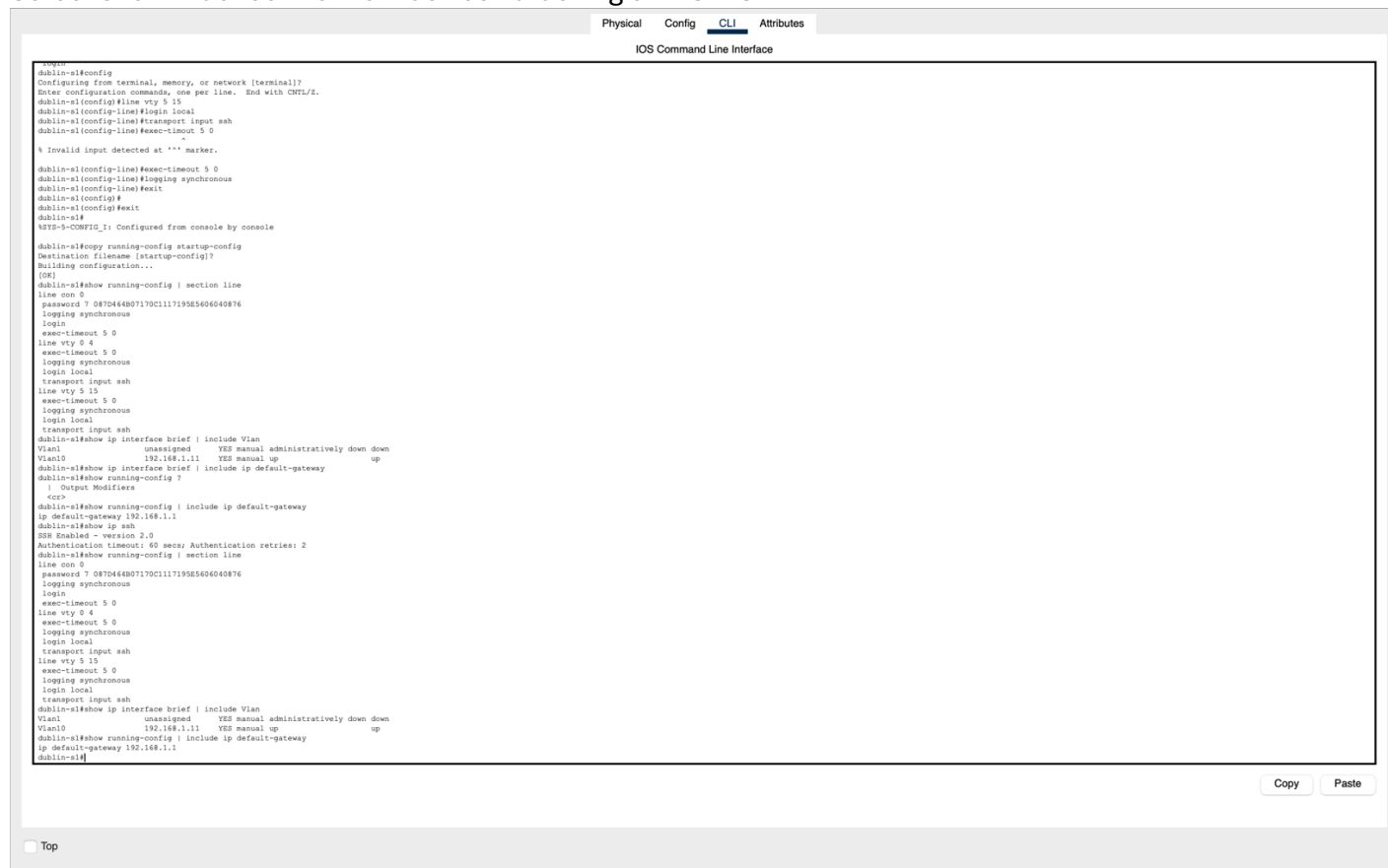
I hardened the routers and switches mainly by protecting how they are managed. Telnet was not used because it sends data in plain text, so I used SSH version 2 instead. The VTY lines were set to accept SSH only and session timeouts were added so an unused admin session would not stay open forever. I also used the London AAA server at 192.168.3.7 for RADIUS login. The technician accounts were used for normal admin access, while localadmin was kept as a backup if the AAA server was unavailable. Commands such as `aaa new-model`, `group radius local`, `ip ssh version 2` and `transport input ssh` were used. This was important because if someone gets control of a router or switch, they could change routes, VLANs or ACLs and affect the whole network.

Commands Used:

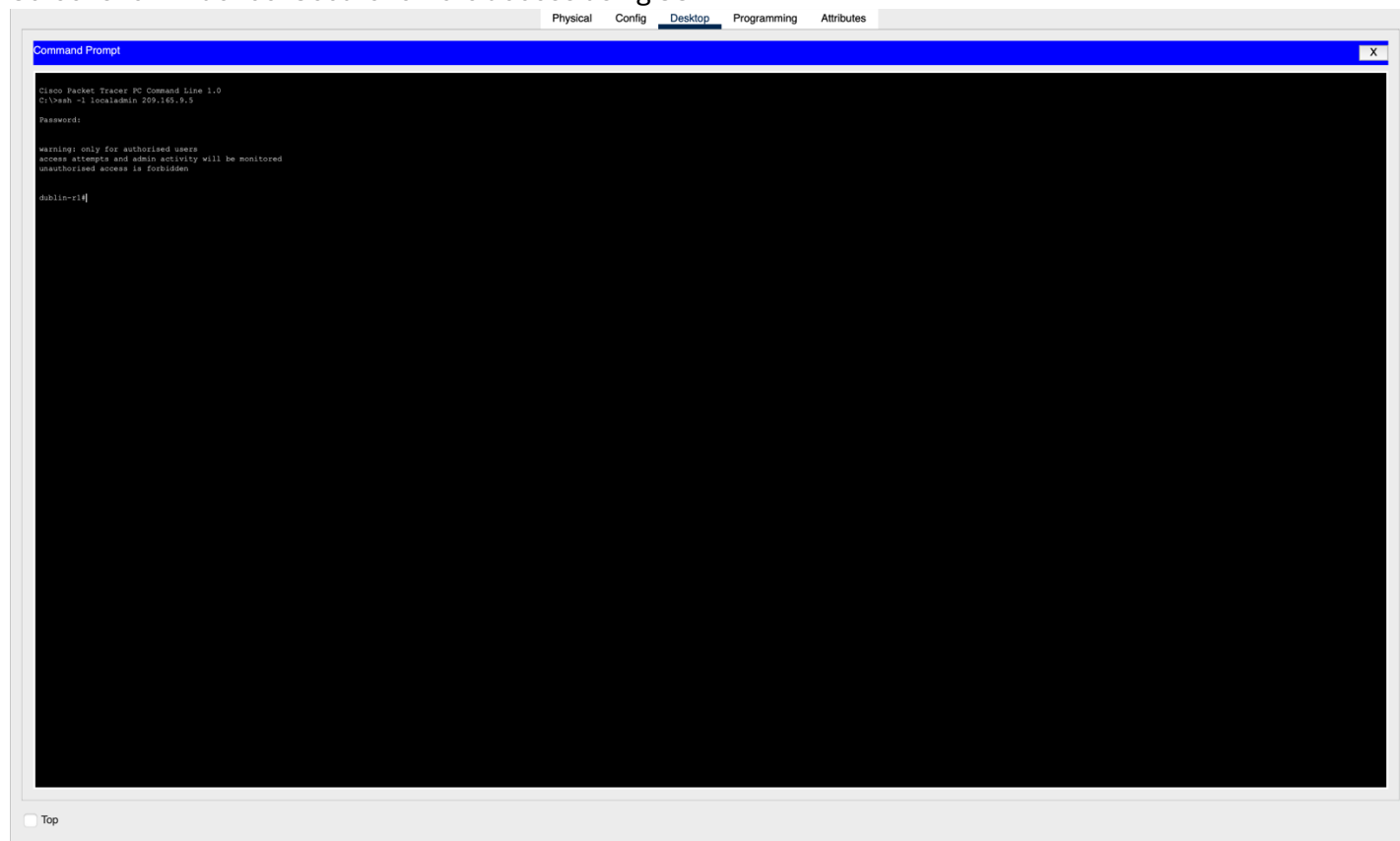
```
aaa new-model
aaa authentication login default group radius local
ip ssh version 2
line vty 0 4
transport input ssh

login authentication default
```

Screenshot Evidence: Network device hardening on DUB-S1



Screenshot Evidence: Secure remote access using SSH



Site-to-Site VPN and Routing Security

I set up a site-to-site IPsec VPN between Dublin and Beijing so traffic between the two sites would be protected while crossing the public network. IKE Phase 1 used AES, SHA, a pre-shared key and Diffie-Hellman group 2. Packet Tracer did not accept the ESP-AES transform syntax I first tried, so I used ESP-3DES with SHA-HMAC for the working transform set. A crypto ACL selected the traffic that should go through the VPN and the crypto map was applied to the public interfaces. I also protected OSPF between Dublin and London with message-digest authentication and the same MD5 key on both serial interfaces. This helps stop fake routing updates and makes it harder to redirect traffic. I checked the VPN, OSPF and BGP after the changes to make sure the routing still worked.

Commands Used:

```
crypto isakmp
  encr aes
  hash sha
  authentication pre-share
  group 2
crypto ipsec transform-set vpn-set esp-3des esp-sha-hmac
interface serial 0/0/0
  ip ospf authentication message-digest
  ip ospf message-digest-key 1 md5 <key>
```

Screenshot Evidence: Site-to-Site VPN configuration on Beijing-R1

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
beijing-r1#show crypto isakmp policy
Global IKE policy
Protection suite of priority 10
  encryption algorithm: AES - Advanced Encryption Standard (128 bit keys)
  hash algorithm:      Secure Hash Standard
  authentication method: Pre-Shared Key
  Diffie-Hellman group: #2 (1024 bit)
  lifetime:             86400 seconds, no volume limit
beijing-r1#cp
Translating "cp"
% Unknown command or computer name, or unable to find computer address

beijing-r1#show crypto ispec transform-set
% Invalid input detected at *** marker.

beijing-r1#show crypto ispec transform-set
% Invalid input detected at *** marker.

beijing-r1#show crypto ispec transform-set
Transform set vpn-set: ( esp-3des esp-sha-hmac )
  will negotiate = ( Tunnel, 1,
Transform set #1(default_transform_set_1) ( esp-aes esp-sha-hmac )
  will negotiate = ( Transport, 1,
Transform set #1(default_transform_set_0) ( esp-3des esp-sha-hmac )
  will negotiate = ( Transport, 1,

beijing-r1#conf t
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
beijing-r1(config)#crypto map vpmap 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
beijing-r1(config-crypto-map)#
beijing-r1(config-crypto-map)#description Beijing-to-dublin
beijing-r1(config-crypto-map)#set peer 150.10.1.1
beijing-r1(config-crypto-map)#set transform-set vpn-set
beijing-r1(config-crypto-map)#match address vpm-traffic
beijing-r1(config-crypto-map)#exit
beijing-r1(config)#
beijing-r1(config)#interface gigabitEthernet 0/0
beijing-r1(config-if)#crypto map vpmap
*Can 3 0716104780: NCPVPO-6-ISAKMP_ON_OFF: ISAKMP is ON
beijing-r1(config-if)#exit
beijing-r1(config)#end
beijing-r1#
*Jul 28, 20:05:07.055: SYS-5-CONFIG_I: Configured from console by console
beijing-r1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
beijing-r1#show crypto map
Crypto Map vpmap 10 ipsec-isakmp
  Peer = 150.10.1.1
  Extended IP access list vpm-traffic
  access-list vpm-traffic permit ip 192.168.5.0 0.0.0.255 192.168.1.0 0.0.0.255
  access-list vpm-traffic permit ip 192.168.5.0 0.0.0.255 192.168.2.0 0.0.0.255
  access-list vpm-traffic permit ip 192.168.5.0 0.0.0.255 192.168.3.0 0.0.0.255
  access-list vpm-traffic permit ip 192.168.5.0 0.0.0.255 192.168.4.0 0.0.0.255
  Current peer: 150.10.1.1
  Security association lifetime: 4608000 kilobytes/3600 seconds
  PFS (Y/N): N
  Transform set=
    Vpn-set,
  Interfaces using crypto map vpmap:
    GigabitEthernet0/0
beijing-r1#
```

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Screenshot Evidence: OSPF routing authentication between Dublin – London Router

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
*Hello due in 00:00:00
Index 3/3, flood queue length 0
Next 0x0/0/0x0/0
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
Adjacent with neighbor 209.165.9.6
Suppress hello for 0 neighbor(s)
beijing-r1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES NVRAM up up
GigabitEthernet0/0.10 192.168.1.1 YES manual up up
GigabitEthernet0/0.20 192.168.2.1 YES manual up up
GigabitEthernet0/1 unassigned YES NVRAM administratively down down
Serial0/0/0 209.165.9.5 YES NVRAM up up
Serial0/0/1 unassigned YES NVRAM down down
GigabitEthernet0/1/0 150.10.1.1 YES NVRAM up up
Vlan1 unassigned YES unset administratively down down
beijing-r1#
beijing-r1#conf t
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
beijing-r1(config)#interface serial 0/0/0
beijing-r1(config-if)#ip ospf message-digest-key 1 md5 087122
% Invalid input detected at *** marker.

beijing-r1(config-if)#ip ospf message-digest-key 1 md5 087122
beijing-r1(config-if)#exit
beijing-r1(config)#
beijing-r1#
*Jul 30, 23:46:18.4646: SYS-5-CONFIG_I: Configured from console by console
beijing-r1#conf t
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
beijing-r1(config)#interface serial 0/0/0
beijing-r1(config-if)#ip ospf authentication message-digest
beijing-r1(config-if)#exit
beijing-r1(config)#end
beijing-r1#
*Jul 30, 23:49:18.4949: SYS-5-CONFIG_I: Configured from console by console
beijing-r1#
beijing-r1#
*Jul 30, 23:49:51.4949: 23:49:51: VOSPF-5-ADJCHG: Process 1, Nbr 209.165.9.6 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired
*Jul 30, 23:49:51.4949: 23:49:51: VOSPF-5-ADJCHG: Process 1, Nbr 209.165.9.6 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
*Jul 30, 23:49:51.4949: 23:49:51: VOSPF-5-ADJCHG: Process 1, Nbr 209.165.9.6 on Serial0/0/0 from LOADING to FULL, Loading Done
beijing-r1#
*Jul 30, 23:49:59.4949: 23:49:59: VOSPF-5-ADJCHG: Process 1, Nbr 209.165.9.6 on Serial0/0/0 from LOADING to FULL, Loading Done
beijing-r1#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
209.165.9.6 0 FULL/ - 00:00:31 209.165.9.6 Serial0/0/0
beijing-r1#show ip ospf interface serial 0/0/0

Serial0/0/0 is up, line protocol is up
Internet address is 209.165.9.5/30, Area 0
Process ID 1, Router ID 209.165.9.5, Network Type POINT-TO-POINT, Cost: 64
Transmit Delay is 1 sec, State POINT-TO-POINT,
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
Hello due in 00:00:00
Index 3/3, flood queue length 0
Next 0x0/0/0x0/0
Last flood scan length is 1, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 1, Adjacent neighbor count is 1
Adjacent with neighbor 209.165.9.6
Suppress hello for 0 neighbor(s)
Message digest authentication enabled
Your neighbor key id is 1
beijing-r1#
```

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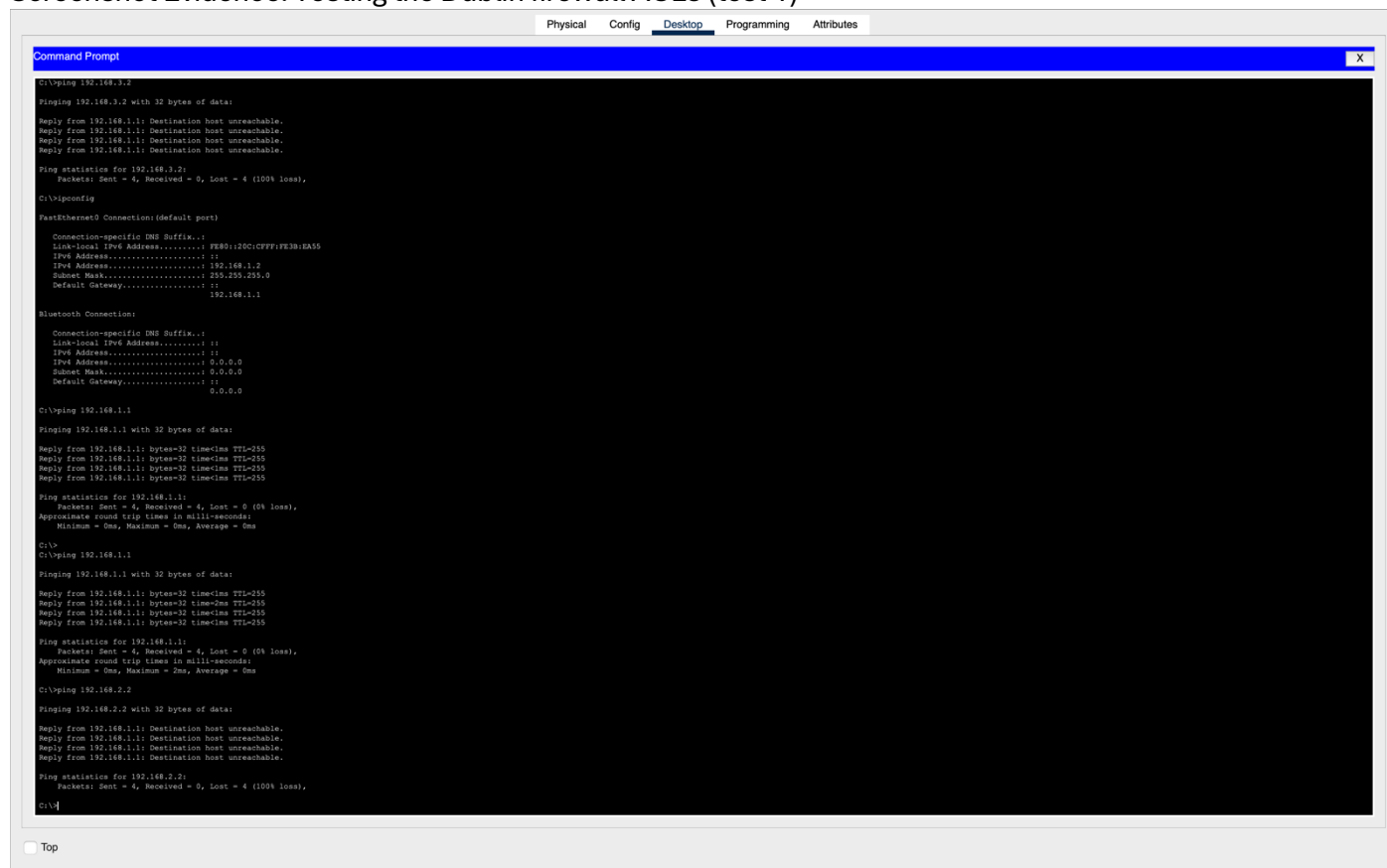
Firewall and Access Control Lists

I used extended ACLs on DUBLIN-R1 as a simple router-based firewall. Sales and Purchasing were blocked from directly reaching each other. Beijing was allowed to reach the Purchasing network, but it was blocked from Sales and the London Admin network. Beijing was also only allowed to use selected London services such as DNS, Web, FTP, NTP and Syslog. The ACLs were placed on the router interfaces in the direction where the traffic needed to be checked. For example, SALES-IN was applied inbound on G0/0.10 and PURCHASING-IN inbound on G0/0.20. The order of the ACL lines mattered because Cisco checks them from top to bottom. DHCP traffic had to be permitted before the deny rules so Purchasing could still get an address. This setup reduces lateral movement because a device in one area cannot automatically access every other part of the network.

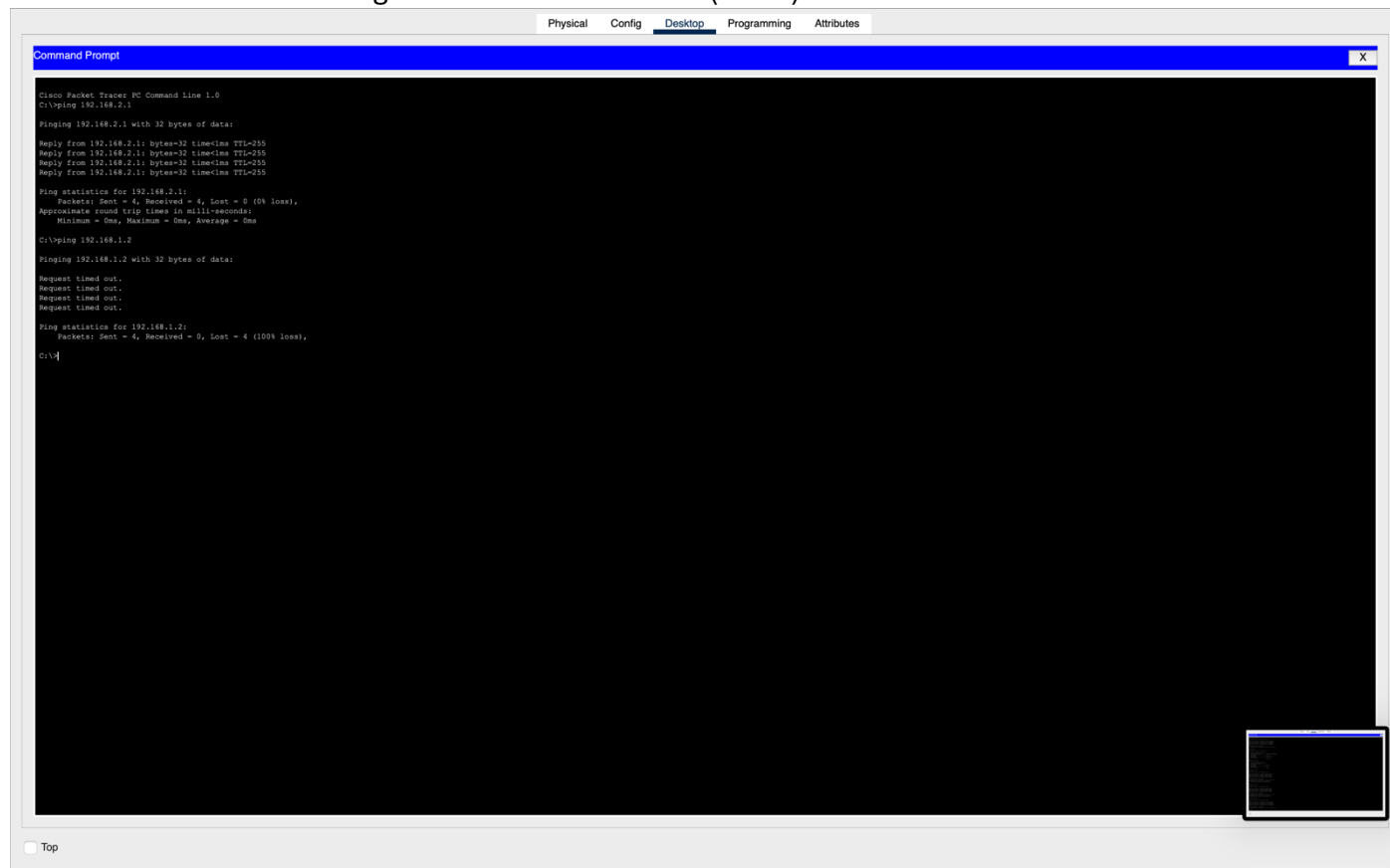
Commands Used:

```
ip access-list extended sales-in
  permit udp any eq 68 any eq 67
deny ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
  permit ip 192.168.1.0 0.0.0.255 any
ip access-list extended sales-protect
deny ip 192.168.5.0 0.0.0.255 192.168.1.0 0.0.0.255
  permit ip any any
```

Screenshot Evidence: Testing the Dublin firewall ACLs (test 1)



Screenshot Evidence: Testing the Dublin firewall ACLs (test 2)



Monitoring: NTP and Syslog

For monitoring, I used the NTP server at 192.168.3.5 and the Syslog server at 192.168.3.6. The routers and switches were pointed to these servers with `ntp server` and `logging` commands. NTP is useful because logs are much easier to compare when every device has the correct time. Syslog keeps messages in one central place instead of leaving them only on each device. This would make troubleshooting and investigating a security problem much easier.

Layer 2 Switch Security

I also secured the switches because attacks can happen before traffic ever reaches a router ACL. User and server ports were set as access ports and port security was used with sticky MAC learning, a maximum of one MAC address and restrict as the violation mode. PortFast and BPDU Guard were only used on end-device ports. The switch trunks were configured manually and VLAN 999 was used as the native and parking VLAN. Unused ports were moved to VLAN 999 and shut down. DHCP snooping was enabled on Dublin VLANs 10 and 20, with only the real DHCP server port and required uplinks trusted. Before changing ports, I checked `show interfaces status`, `CDP neighbours` and trunk information so I did not shut down a real link by mistake. These controls help stop rogue devices, fake DHCP servers and simple Layer 2 attacks.

Commands Used:

```
switchport mode access
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
spanning-tree portfast
spanning-tree bpduguard enable
ip dhcp snooping
ip dhcp snooping vlan 10,20
```

Screenshot Evidence: Layer 2 switch security on London-S4

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
*Aug 01, 03:50:39.5050: STS-5-COMPLETE: Configured from console by console

london-s4#show int status
Port      Name      Status      Vlan      Duplex  Speed Type
Fa0/1     dns-server connected    1          a-full  a-100  10/100BaseTX
Fa0/2     web-server connected    1          a-full  a-100  10/100BaseTX
Fa0/3     ftp-server  connected    1          a-full  a-100  10/100BaseTX
Fa0/4     rtp-server  connected    1          a-full  a-100  10/100BaseTX
Fa0/5     syslog-server connected    1          a-full  a-100  10/100BaseTX
Fa0/6     asa-server  connected    1          a-full  a-100  10/100BaseTX
Fa0/7     unused      disabled 999        auto    auto   10/100BaseTX
Fa0/8     unused      disabled 999        auto    auto   10/100BaseTX
Fa0/9     unused      disabled 999        auto    auto   10/100BaseTX
Fa0/10    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/11    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/12    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/13    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/14    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/15    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/16    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/17    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/18    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/19    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/20    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/21    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/22    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/23    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/24    unused      disabled 999        auto    auto   10/100BaseTX
Gig0/1    london-r1-server-1 connected    1          a-full  a-100  10/100/1000BaseTX
Gig0/2    unused      disabled 999        auto    auto   10/100/1000BaseTX

london-s4#show vlan brief

VLAN Name      Status      Ports
----
1    default    active      Fa0/1, Fa0/2, Fa0/3, Fa0/4
999  parking-lot active      Fa0/5, Fa0/6, Gig0/1
1002 fddi-default active
1003 token-ring-default active
1004 fddiint-default active
1005 trnet-default active

london-s4#show port-security
Secure Port MacSecureAddr CurrentAddr SecurityViolation Security Action
(Count)      (Count)      (Count)
-----
Fa0/1        1          1          0          Restrict
Fa0/2        1          1          0          Restrict
Fa0/3        1          1          0          Restrict
Fa0/4        1          1          0          Restrict
Fa0/5        1          1          0          Restrict
Fa0/6        1          1          0          Restrict

london-s4#show port-security address
Secure Mac Address Table
-----
Vlan  Mac Address      Type      Ports      Remaining Age (mins)
----
1     0090.21C9.0928   SecureSticky Fa0/1      -
1     0004.8A4D.0249   SecureSticky Fa0/2      -
1     0005.5E3D.C242   SecureSticky Fa0/3      -
1     0004.8A4B.9861   SecureSticky Fa0/4      -
1     000A.4187.4132   SecureSticky Fa0/5      -
1     0080.F778.254D   SecureSticky Fa0/6      -

Total Addresses in System (excluding one mac per port) : 0
Max Addresses limit in System (excluding one mac per port) : 1024

london-s4#
```

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Screenshot Evidence: Layer 2 switch security on DUB-S3 (also trunking and unused-port security)

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
Fa0/1     Desg PWD 19      128.1    P2p
Fa0/2     Desg PWD 19      128.2    P2p
Gig0/1    Desg PWD 4        128.25   P2p

dublin-s3#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    999
Fa0/2     on        802.1q         trunking    999
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/1     10,20,999
Fa0/2     10,20,999
Gig0/1     1-1005

Port      Vlans allowed and active in management domain
Fa0/1     10,20,999
Fa0/2     10,20,999
Gig0/1     1,10,20,999

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,999
Fa0/2     10,20,999
Gig0/1     1,10,20,999

dublin-s3#show interfaces status
Port      Name      Status      Vlan      Duplex  Speed Type
Fa0/1     trunk-to-dub1 connected trunk    a-full  a-100  10/100BaseTX
Fa0/2     trunk-to-dub2 connected trunk    a-full  a-100  10/100BaseTX
Fa0/3     connected 10          a-full  a-100  10/100BaseTX
Fa0/4     unused      disabled 999        auto    auto   10/100BaseTX
Fa0/5     unused      disabled 999        auto    auto   10/100BaseTX
Fa0/6     unused      disabled 999        auto    auto   10/100BaseTX
Fa0/7     unused      disabled 999        auto    auto   10/100BaseTX
Fa0/8     unused      disabled 999        auto    auto   10/100BaseTX
Fa0/9     unused      disabled 999        auto    auto   10/100BaseTX
Fa0/10    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/11    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/12    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/13    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/14    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/15    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/16    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/17    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/18    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/19    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/20    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/21    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/22    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/23    unused      disabled 999        auto    auto   10/100BaseTX
Fa0/24    unused      disabled 999        auto    auto   10/100BaseTX
Gig0/1    connected trunk    a-full  a-100  10/100/1000BaseTX
Gig0/2    notconnect 1          auto    auto   10/100/1000BaseTX

dublin-s3#show vlan brief

VLAN Name      Status      Ports
----
1     default    active      Gig0/2
10    Sales     active      Fa0/3
20    Purchasing active      Fa0/4, Fa0/5, Fa0/6, Fa0/7
999  parking-lot active      Fa0/8, Fa0/9, Fa0/10, Fa0/11
1002 fddi-default active
1003 token-ring-default active
1004 fddiint-default active
1005 trnet-default active

dublin-s3#
```

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NAT and PAT

PAT was configured on LONDON-R1 for the London Admin network. The LAN interfaces were marked as ip nat inside and Serial0/0/0 was marked as ip nat outside. The LONDON-PAT ACL excluded the company private networks from translation and allowed other Admin traffic to use the Serial0/0/0 address 209.165.9.6 with overload. This lets several private hosts share one public address. I checked it with show ip nat translations and could see the inside local address being translated. Separate static NAT for DNS, Web and FTP was not possible because the /30 public link had no spare usable addresses.

Commands Used:

```
ip nat inside source list london-pat interface serial0/0/0 overload
```

Testing and Verification

I tested the network after each major change and again at the end. I used positive tests for traffic that should work and negative tests for traffic that should be blocked. For example, Sales and Purchasing could reach their own gateways but could not reach each other. Beijing could reach the allowed Purchasing network, while blocked destinations failed. I also tested SSH from the London Admin network, checked OSPF neighbours, VPN status, PAT translations and the monitoring servers. Testing also helped me find mistakes. At one point the wrong ACL was applied to Purchasing, and a native VLAN mismatch caused a trunk problem. I fixed each issue and retested before saving. This was useful because a command can look correct on screen but still be on the wrong interface or in the wrong direction.

Test	Expected Result	Evidence
Sales – Purchasing	Blocked	ACL deny counter increases
Purchasing – gateway/DHCP	Allowed	192.168.2.0 lease and successful ping
Beijing – Purchasing	Allowed	Successful end-host ping through VPN
London Admin – routers	Allowed	RADIUS/SSH login succeeds
London Admin – public test IP	Allowed	PAT translation appears
OSPF/VPN/BGP	Allowed	FULL/QM_IDLE/established

Other Security Areas to Consider

In a real company I would add more security than Packet Tracer shows here. PCs and servers should have anti-malware or EDR, host firewalls, regular patching and secure backups. Admin accounts should use multi-factor authentication. I would also consider 802.1X so users or devices have to authenticate before getting normal network access. Wireless networks should use WPA2/WPA3-Enterprise with RADIUS. An IDS/IPS and a SIEM would help detect and investigate attacks, while locked network rooms would protect the physical equipment. These controls matter because network ACLs cannot stop every threat, especially malware already running on an allowed device.

Limitations and Lessons Learned

Packet Tracer does not support every IOS command exactly like real equipment. ESP-AES syntax was not available for the VPN transform set, switchport nonegotiate was rejected and some show running-config filters did not work. VTP client mode also affected where VLAN 999 could be created. The main lesson for me was to make one change at a time, test it, and only save once I knew the network was still working.

Conclusion

Overall, the network now has several security layers working together. SSH and AAA protect management access, IPsec protects site traffic, OSPF authentication protects routing, ACLs control who can reach different networks, NTP and Syslog support monitoring, Layer 2 controls protect switch access and PAT handles London Admin traffic going outside. The final tests included both traffic that should succeed and traffic that should fail. This showed that the security rules were actually working and not just present in the configuration.

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